

Overview

The ANC300 class enables communication and control of the ANC300 device, providing functions to set frequencies and voltages, move axes, and configure operational modes. To establish a connection, it is essential to know the COM port the ANC300 is connected to. You can find this in the Windows Device Manager under "Ports (COM & LPT)".

Requirements

1. The ANC300 device must be connected to the PC via RS232 with the correct COM port specified or USB connection.
2. You need to find the COM port of the ANC300 device through **Device Manager** under **Ports (COM & LPT)**.

Class Definition

```
1. classdef ANC300 < handle
2.     properties
3.         device % Serial connection object for the ANC300
4.     end
5.     properties (Constant)
6.         baudrate = 38400 % Baudrate set according to ANC300 specifications
7.     end
```

Constructor

```
1. function obj = ANC300(port)
2.     % Constructor: Initializes a serial connection with the specified COM port.
3.     % Args:
4.     % port (str): The COM port identifier (e.g., 'COM3').
5.
6.     obj.device = serialport(port, obj.baudrate);
7.     obj.device.DataBits = 8;
8.     obj.device.StopBits = 1;
9.     obj.device.Parity = "none";
10.    obj.device.Timeout = 10;
11.    obj.device.FlowControl = "none";
12.    configureTerminator(obj.device, "CR/LF");
13. end
```

Explanation: Creates a serial connection using a specified COM port. Configures communication parameters to match ANC300 requirements, setting the terminator to CR/LF for command completion.

Methods

1. Set Frequency

function response = set_frequency(obj, Hz, axis)

- **Purpose:** Sets the stepping frequency for a specified axis.
- **Args:**
 - Hz (double): Frequency in Hz.

- axis (int): Axis identifier.
- **Returns:** Response from the device.

2. Get Frequency

function [cmd, freq] = get_frequency(obj, axis)

- **Purpose:** Retrieves the stepping frequency for a specified axis.
- **Args:**
 - axis (int): Axis identifier.
- **Returns:**
 - cmd (str): Command confirmation message.
 - freq (str): Frequency in Hz.

3. Set Voltage

function set_voltage(obj, voltage, axis)

- **Purpose:** Sets the voltage for a specified axis.
- **Args:**
 - voltage (double): Desired voltage.
 - axis (int): Axis identifier.

4. Step Up

function stepup(obj, steps, axis)

- **Purpose:** Moves the axis a specified number of steps upwards.
- **Args:**
 - steps (int): Number of steps to move.
 - axis (int): Axis identifier.

5. Step Down

function stepdown(obj, steps, axis)

- **Purpose:** Moves the axis a specified number of steps downwards.
- **Args:**
 - steps (int): Number of steps.

- `axis (int)`: Axis identifier.

6. Get Voltage

function [cmd, volt, other] = get_voltage(obj, axis)

- **Purpose:** Retrieves the current voltage for a specified axis.
- **Args:**
 - `axis (int)`: Axis identifier.
- **Returns:**
 - `cmd (str)`: Command response.
 - `volt (str)`: Voltage value.
 - `other (str)`: Additional response information.

7. Stop

function response = stop(obj, axis)

- **Purpose:** Stops any motion for a specified axis.
- **Args:**
 - `axis (int)`: Axis identifier.
- **Returns:** Response from the device.

8. Set Mode

function set_mode(obj, mode, axis)

- **Purpose:** Configures the operational mode of the specified axis.
- **Args:**
 - `mode (str)`: Mode identifier (e.g., "gnd", "cap", "stp").
 - `axis (int)`: Axis identifier.

Usage Example

```

1. % Initialize ANC300 on COM3
2. controller = ANC300('COM3');
3.
4. % Set frequency to 500 Hz for axis 1
5. controller.set_frequency(500, 1);
6.
7. % Move axis 1 up by 100 steps
8. controller.stepup(100, 1);
9.
10. % Get the voltage for axis 1
11. [cmd, volt, ~] = controller.get_voltage(1);
12. disp(['Voltage for axis 1: ', volt]);

```

Note: The flush method is used to clear the device's input and output buffer, preventing old data from interfering with current commands.

ANC300 GUI

Make sure the Class ANC300.m file is in the same directory as the GUI file ANC300_GUI, to function properly.

Calling ANC300_GUI:

```

1. clear
2. clc
3. app = ANC300_GUI(Port)

```

With specified port as mentioned above, which can be found in device manager. You can have as many GUI's if you input the corresponding port and new app variable.

The UI should look like this:

	Frequency(Hz)	Voltage(V)	Mode
X-Axis	500	35	Gnd
Y-Axis	510	35	Gnd
Z-Axis	500	35	Gnd

Update

Out Z ● Out Y ● Out X ●

Step up Z Step up Y Step up X

Step down Z Step down Y Step down X

Set frequency and voltage for the piezo, as well as its mode which is within the dropdown menu.

Make sure to press update after setting appropriate values.

You can move the piezo using step up buttons on the right, which moves the piezo one step at a time.

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